Exploring Hybrid Contrastive Learning and Scene-to-Label Information for Multilabel Remote Sensing Image Classification

Tiecheng Song, *Member, IEEE*, Shufen Bai, Feng Yang, Chenqiang Gao, Haonan Chen, *Senior Member, IEEE*, and Jun Li, *Fellow, IEEE*

***Abstract*— Multilabel remote sensing (RS) image classification aims to predict multiple semantic labels from an RS image. Previous methods [e.g., graph convolution networks (GCNs)] focus on mining the relationships of multiple labels, neglecting that the scene information is closely related to labels. To remedy this deficiency, in this article we propose a novel end-to-end deep neural network for multilabel RS image classification. In the proposed network, we use the GCN as the base model and introduce several new components to improve the classification performance. First, we explore hybrid contrastive learning (CL), including supervised transformation-based CL and unsupervised mix-based CL, to explicitly learn discriminative scene representations. Then, we apply the GCN-based classifier to the learned scene representations to obtain initial label prediction scores. Meanwhile, we pass the scene representations to a softmax layer to predict the probability that each image belongs to each specific scene class and use the scene-to-label information with the law of total probability to calibrate the initial label prediction scores. Finally, we incorporate CL, scene classification, and multilabel classification into a unified learning framework using uncertainty to weigh different losses. Experimental results on two benchmark RS datasets demonstrate the superiority of our proposed network for multilabel image classification.**

***Index Terms*— Contrastive learning (CL), graph convolution, multilabel image classification, multitask learning, remote sensing (RS), scene classification.**

Manuscript received 4 October 2023; revised 10 April 2024; accepted 26 June 2024. Date of publication 2 July 2024; date of current version 15 July 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFA1004100; in part by the National Natural Science Foundation of China under Grant 62371084, Grant 62371083, and Grant 62176035; in part by the Natural Science Foundation of Chongqing, China, under Grant CSTB2022NSCQ-MSX1418; in part by China Postdoctoral Science Foundation under Grant 2022MD723727 and Grant 2021MD703940; in part by the Special Support for Chongqing Postdoctoral Research Project under Grant 2022CQBSHTB2041 and Grant 2021XM3036; and in part by the Funding of the Institute for Advanced Sciences, Chongqing University of Posts and Telecommunications, under Grant E011A2022330. *(Corresponding author: Chenqiang Gao.)*

Tiecheng Song is with the School of Communications and Information Engineering and the Institute for Advanced Sciences, Chongqing University of Posts and Telecommunications, Chongqing 400065, China.

Shufen Bai and Feng Yang are with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China.

Chenqiang Gao is with the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen 518107, China (e-mail: gaochq6@mail.sysu.edu.cn).

Haonan Chen is with the Department of Electrical and Computer Engineering, Colorado State University, Fort Collins, CO 80523 USA.

Jun Li is with the School of Computer Science and Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China.

Digital Object Identifier 10.1109/TGRS.2024.3422031

NOMENCLATURE

Abbreviations	Full definitions
RS	Remote sensing.
GCN	Graph convolution network.
CL	Contrastive learning.
CNN	Convolutional neural network.
RNN	Recurrent neural network.
LSTM	Long short-term memory.
GCN	Graph convolutional network.
HCL	Hybrid contrastive learning.
SLP	Scene-to-label prediction.
SIFT	Scale-invariant feature transform.
LBP	Local binary pattern.
UAV	Unmanned aerial vehicle.
SSVM	Structured support vector machine.
ML-GCN	GCN for multilabel image classification.
MLP	Multilayer perceptron.
FC	Fully connected layer.
ReLU	Rectified linear unit.
VGG	Visual geometry group.
ResNet	Residual network.
GloVe	Global vectors for word representation.
MoCo	Momentum contrast.
SimCLR	Simple framework for contrastive learning of visual representations.
Grad-CAM	Gradient-weighted class activation mapping.
FLOPs	Floating-point operations.

I. INTRODUCTION

WITH the development of aerial photography and satellite technologies, RS images have been applied in a variety of tasks, including land cover recognition [1], change detection [2], object detection [3], and scene classification [4], [5]. Among these tasks, RS image scene classification aims to classify an RS image into a single scene class. However, because high-resolution RS images usually contain multiple types of semantic information, the classification results indicating one single scene class cannot accurately reflect the image's content. To this end, multilabel image classification [6] is developed, which aims to find one or more semantic/object labels present in an image. Its main challenge lies in that the size of output label space grows exponentially as the number

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

of labels increases, and the exploitation of label correlations information becomes crucial for addressing this challenge [6].

In the literature, many multilabel classification methods, especially deep-learning-based ones, have been proposed by considering the semantic correlation between labels. For example, the works by Wang et al. [7] and Liu et al. [8] use the CNN to extract features and use the RNN or the bidirectional LSTM network to capture label correlations. The methods [9], [10], [11], [12], [13], [14], [15], [16] use target localization techniques or attention mechanisms to find semantic label regions for multilabel classification. The methods [17], [18], [19], [20], [21], [22], [23] model the label relationship as a graph. Among them, Chen et al. [17] explored word embeddings and co-occurrence probabilities between labels for graph learning and applied the GCN-based classifier to input images for multilabel prediction. Using both visual and text information, the method in [17] is a flexible and promising learning framework for multilabel classification. However, it neglects that the scene information is closely related to labels (e.g., the airport scene is related to labels like airplane and runway; the beach scene is related to sea and sand) and the guidance effects of scenes on label inference are unexplored to improve the multilabel classification performance. For high-resolution RS images with rich semantic information, how to build effective scene representations and subsequently leverage scenes to guide label classification remains a challenging problem.

To address the above problem, in this article we propose a novel deep network by exploring hybrid CL and scene-to-label information (HS) for multilabel RS image classification. The proposed network (called HSNet) uses the GCN-based multilabel classifier introduced in [17] to obtain the initial label prediction scores and further introduces several novel components to improve the performance, including scene representations, SLP, and the optimization of loss functions. First, HSNet explores hybrid CL to explicitly learn discriminative scene representations and improve the model's generalization ability. Second, HSNet learns to predict the probability that each image belongs to each specific scene class and uses the prior information of scene-to-label to calibrate the initial label prediction scores. Third, HSNet incorporates CL, scene classification, and multilabel classification into a unified learning framework where uncertainty is used to weigh multiple losses and balance different tasks. Experiments on two benchmark RS datasets show the superiority of HSNet for multilabel image classification.

The contributions of this article are summarized as follows.

1) We propose a novel GCN-based deep network model for multilabel RS image classification, which incorporates CL, scene classification and multilabel classification into a unified learning framework.
2) We integrate data augmentation into CL and present an HCL module, including supervised transformation-based CL and unsupervised mix-based CL, to explicitly model scene representations and learn discriminative scene features.
3) We design an SLP module. In SLP, we use a softmax layer to predict the probability that each image belongs to each specific scene class and use the scene-to-label information with the law of total probability to calibrate the initial label prediction scores obtained from the GCN classifier. The resulting GCN classifier and SLP module are jointly trained to boost each other.
4) We take advantage of uncertainty to learn to weigh different losses for multitask learning, which improves the model's classification performance while lightening the manual tuning of weight hyperparameters.

The rest of this article is organized as follows. The related work is presented in Section II. The proposed network is introduced in Section III. The experiments are conducted in Section IV. Finally, the conclusion is drawn in Section V. The abbreviations used in this article are listed in Nomenclature.

II. Related Work

A. RS Image Scene Classification

RS image scene classification aims to assign one single image-level label (i.e., scene class) to an input image, and three types of image features, i.e., low-level, mid-level, and high-level ones, are generally used for this task. Low-level features are often handcrafted descriptors used to represent local or global information about color, texture, shape, etc. The classic works include Gabor filter, SIFT [24], and LBP [25]. Mid-level features encode the image's higher order statistical information or semantic relationship, and the representative works are the bag-of-visual-words model [26] and the probabilistic topic model [27]. High-level features are typically derived from deep learning methods such as CNNs and transformers, which have become the mainstream techniques for discriminative representation learning. For example, Yuan et al. [28] used the VGG-19 network [29] to extract global and local features for RS image scene classification. Zheng et al. [30] proposed a multilayer and multiattention fusion network based on ResNet-50 [31]. Wang et al. [32] combined CNN and LSTM to build an attentional recurrent convolutional network. Tang et al. [33] combined CNN and transformer to develop a multiscale transformer and cross-level attention learning model. Xu et al. [34] used vision transformer as the teacher model to guide the learning of a small student network ResNet-18 [31] for RS scene classification.

B. Multilabel Classification

Different from single-label scene classification, multilabel classification aims to assign one or more semantic labels (i.e., object classes) to an input image, which is a practical and challenging task in the RS field because high-resolution RS images usually contain multiple objects. Methods based on CNN, CNN + RNN, localization/attention, and graph structure have been developed for multilabel image classification, which are described below.

For the CNN-based methods, Zeggada et al. [35] extracted CNN features from image tiles and used a radial basis function neural network for multilabel classification of UAV images. Koda et al. [36] explored the SSVM for multilabel classification where image tiles are represented as CNN features and the spatial contextual information is incorporated to optimize the SSVM. Stivaktakis et al. [37] used deep CNN architectures

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

along with data augmentation for multilabel land cover scene classification where the sigmoid activation function is used at the output layer to predict label scores. Although the above works extract discriminative CNN features, the multilabel classifier they used ignores or cannot effectively capture potential semantic dependencies between labels.

As for the CNN + RNN-based methods, Wang et al. [7] used CNNs to extract semantic image representations and used RNNs to learn a joint image-label embedding to characterize the semantic label dependency and the image-label relevance. Liu et al. [8] proposed a semantically regularized embedding layer as the interface between CNN and RNN. These methods do not consider the explicit associations between semantic labels and image contents, and hence the spatial context information is not fully explored.

In view of the above issues, some works leverage object detection technique to learn regional or local informative features. For example, Zhang et al. [9] designed a three-branch multilabel classification framework to make full use of both the discriminant information of the object itself and the spatial contextual information around the object. Zhang et al. [10] explored a fully convolutional localization architecture to locate regions that may contain multiple highly dependent labels and used LSTMs to capture label dependencies at the regional level. The above localization models require a large number of proposals and object-level supervised information. To overcome such problems, some works use attention mechanisms to find semantic object regions. For example, Wang et al. [11] introduced a spatial transformer layer to locate attentional regions in a region-proposal-free way and used an LSTM unit to predict the labels on located regions. Zhu et al. [12] proposed a spatial regularization network based on the attention mechanism to capture spatial and semantic relations of labels with only image-level supervisions. For multilabel RS image classification, Hua et al. [13] proposed a class attention learning layer to capture discriminative class-specific features and used a bidirectional LSTM subnetwork to model the underlying class dependency. In [14], they also used a learnable spatial transformer to localize attentional regions and constructed a label relational inference module to predict multiple object labels of aerial images. Tan et al. [15] used class activation maps to locate semantic attentional regions and explored the transformer to build the relationship between categories for multilabel RS image classification. Bashmal et al. [16] used the visual transformer to address multilabel classification of UAV imagery. They explored the self-attention mechanism to derive the correlation between different objects and mounted a cross-attention module to detect the cross correlation between the image and its augmented view.

Currently, a considerable number of works use the graph structure to mine label relationships and solve multilabel classification problem. For example, Chen et al. [17] constructed a directed graph over the representations of object labels (i.e., word embeddings) and learned stacked GCNs over the label graph to map these label representations into a set of interdependent object classifiers for multilabel image recognition. Chen et al. [18] designed a semantic decoupling module to learn category-related image features and constructed a graph to correlate these features and explore their interactions via a graph propagation mechanism. For multilabel RS image classification, Li et al. [19] used superpixel regions to construct the scene graph where nodes were represented by deep CNN features and designed a multilayer integration graph attention network for multilabel classification. Kang et al. [20] proposed a graph relational network to model the relationship between images and labels by considering a scalable neighbor discriminative loss. Lin et al. [21] proposed a multilabel concept graph to describe the semantic correlations and generate label-level features. In [22], they further designed three grouping mechanisms to obtain image features, which were combined with the concept graph for multilabel classification. Tan et al. [23] leveraged low-rank representation for graph construction in both the feature space and label space and used the constructed two graphs to train a graph-based multilabel classifier.

The proposed multilabel classification network belongs to the graph-based methods, and especially we use the GCN described in [17] for graph learning. The main difference is that previous GCN-based methods neglect that the scene information is closely related to labels and the guiding effects of scene information on label inference are unexplored for multilabel classification. To bridge this gap, in this work we leverage the word-embedding-based GCN classifier [17] to obtain initial label prediction scores and additionally design an SLP module to calibrate these scores. Consequently, the GCN classifier and the SLP module are jointly trained to enhance multilabel classification performance.

C. Contrastive Learning (CL)

CL has recently gained much attention due to its capability of learning meaningful feature representations. The main idea of CL is to construct positive and negative samples relative to an anchor point and optimize a contrastive loss function which encourages the anchor point to be closer to its positive samples and farther from its negative ones. CL can be generally categorized into unsupervised CL and supervised CL.

For unsupervised CL methods, SimCLR [38], MoCo [39] and SimCLRv2 [40] use data augmentation (e.g., image rotation and cropping) to obtain enhanced positive samples for an anchor point, while other samples in the same batch are considered as negative ones. Li et al. [41] proposed a global style and local matching CL network for RS image semantic segmentation. Kim et al. [42] and Wickstrøm et al. [43] used Mixup [44] to mix two samples and extended the concept of CL to semi-positives. Mixup-based CL can learn relative similarities, which reflect how much the mixed images have the original positives. However, the mixed samples are often visually misty and locally unnatural. In addition, without supervised information, the above unsupervised CL methods can increase the distance between two examples from the same class in the learned feature space.

Supervised CL methods [45] explore the class information to pull the features of samples from the same class closer and push those from different classes further away. For example, Huang et al. [46] fused global and local features based on supervised CL for better image representations. Cui et al. [47]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

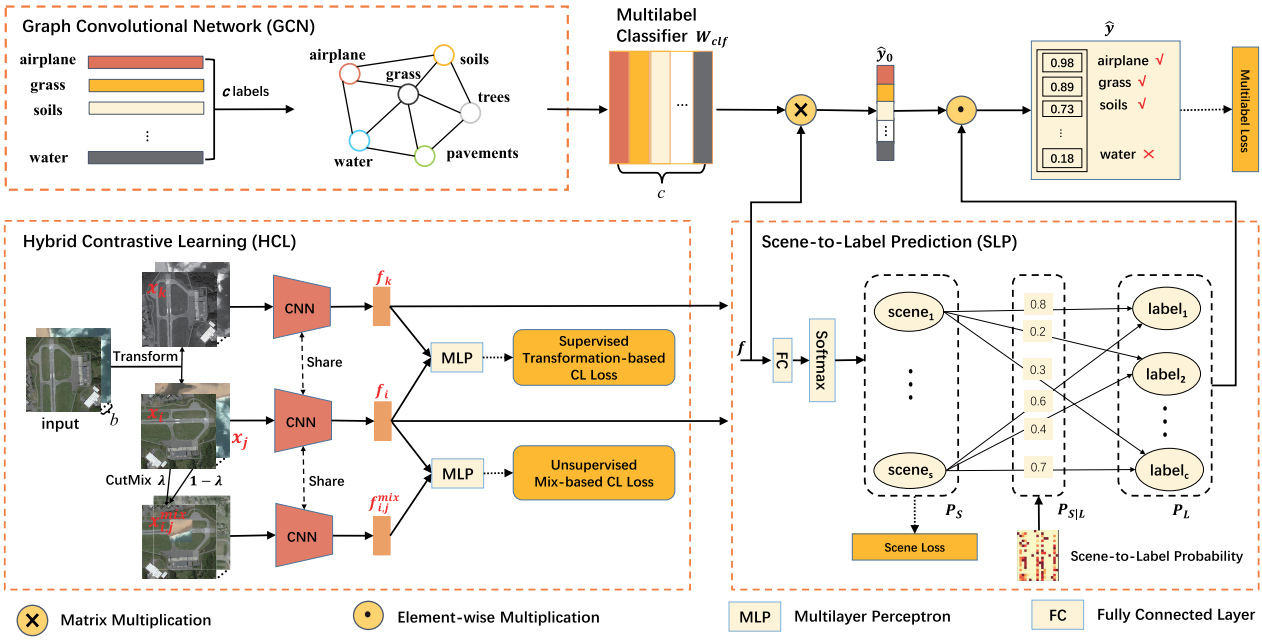


Fig. 1. Pipeline of the proposed HSNet for multilabel RS image classification. In the training phase, given a batch of b images, both the features of transformed and mixed (via CutMix) images are computed via weight-shared CNNs, and only the features of transformed images are fed into the SLP module for subsequent processing. In the inference phase, given a test image, only the feature corresponding to f_i is computed (features corresponding to f_k and f_{ij}^{mix} are no longer used) and fed into the SLP module for multilabel classification.

introduced a set of parametric classwise learnable centers into supervised CL to tackle long-tailed image recognition.

Inspired by the above works, we propose to combine unsupervised and supervised CL to learn discriminative scene representations. Particularly, we introduce CutMix [48], a data augmentation strategy by replacing an image region with a patch from another image, in the unsupervised CL framework to generate diverse images while preserving local naturalness. The learned scene representations will be used for scene classification and also used in the SLP module for multilabel classification. To the best of our knowledge, we are the first to incorporate CL, scene classification, and multilabel classification (via the GCN classifier and the SLP module) into a unified learning framework.

III. Methodology

Fig. 1 shows the pipeline of the proposed HSNet for multilabel RS image classification. First, we put the word vectors and the co-occurrence probabilities of given c labels into GCN to obtain the GCN-based multilabel classifier. Then, we feed the training images into the HCL module to explicitly learn discriminative scene features. After that, we multiply each scene feature by the GCN-based classifier to obtain the initial multilabel prediction scores. Meanwhile, we pass the scene features to the SLP module. In SLP, a softmax layer is used to predict the probability that each image belongs to each specific scene class (this is achieved by a scene classification loss) and subsequently the prior information of scene-to-label is explored to calibrate the initial label prediction scores. The resulting network incorporates CL, scene classification, and multilabel classification into a unified learning framework. Finally, we use uncertainty to weigh different losses for multitask learning. In Section III-A, we first briefly introduce the GCN framework for multilabel classification used in [17], and then detail the other parts of our HSNet.

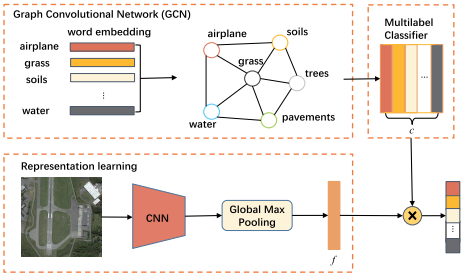


Fig. 2. Framework of ML-GCN proposed in [17] for multilabel classification. The model uses GCN to map c label representations, e.g., word embeddings, to a multilabel classifier (the final output of each GCN node is a classifier corresponding to a specific label), which is applied to the image representation f for multilabel classification.

A. GCN for Multilabel Image Classification (ML-GCN)

ML-GCN [17] builds a directed graph over object labels where each node (label) is represented by word embeddings and updates node representations by propagating information among nodes based on a label correlation matrix. As shown in Fig. 2, the GCN is learned to map the label graph into a set of interdependent label classifiers, which can be applied to an image representation for multilabel classification.

More specifically, the nodes of GCN are word vectors obtained from the GloVe [49] model trained on the Wikipedia dataset. The label correlation matrix is basically constructed by a co-occurrence matrix A whose elements are conditional probability calculated as

$$P_{ij} = P\left(l_j \middle| l_i\right) = N_{ij}/N_i \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

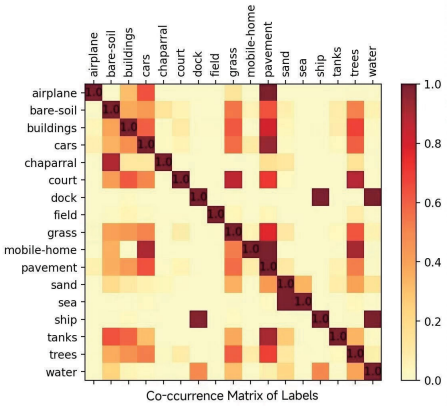


Fig. 3. Co-occurrence matrix of labels for the UCM multilabel RS dataset. This is a conditional probability matrix where the first row shows the occurrence probability of all the possible labels when the label "airplane" appears in one image. The rest rows can be deduced by analogy.

where N_i denotes the occurring times of label l_i in the training set, and N_{ij} denotes the concurring times of l_i and l_j. Fig. 3 shows the co-occurrence matrix of labels for the UC Merced (UCM) multilabel RS dataset. In this work, we adopt the reweighted label correlation matrix, as used in [17], to alleviate the overfitting and oversmoothing problems.

The GCN is learned on the label graph by taking node representations $H^l \in \mathbb{R}^{c\times d}$ and the corresponding correlation matrix $A \in \mathbb{R}^{c\times c}$ as input (where c denotes the number of labels, d denotes the dimensions of word vectors, and l indicates the lth layer for graph convolution) and updates the node representations as $H^{l+1} \in \mathbb{R}^{c\times d'}$, i.e.,

$$H^{l+1} = h\left(\hat{A}H^l W^l\right) \tag{2}$$

where W^l is a transformation matrix to be learned, $\hat{A}$ is the normalized version of A, and $h(\cdot)$ denotes a nonlinear function like LeakyReLU used in [50]. In this work, we learn a two-layer GCN. For the last layer, the output is viewed as a classifier $W_{\text{clf}} \in \mathbb{R}^{c\times d}$, which is directly applied to the image representation $f \in \mathbb{R}^d$ to obtain label prediction scores

$$\hat{y}_0 = W_{\text{clf}} \otimes f \tag{3}$$

where $\hat{y}_0 \in \mathbb{R}^c$, and $\otimes$ denotes matrix multiplication. Based on (3), the whole network can be trained using the multilabel classification loss (i.e., binary cross-entropy loss).

In the following, we will explore CL to learn discriminative image/scene representations and propose using the scene information to calibrate the label prediction scores in (3).

B. Hybrid CL (HCL)

Overfitting caused by insufficient data is a common problem in deep learning. Data augmentation methods, typically containing transformation-based ones (e.g., image rotation and cropping) and mix-based ones (e.g., Mixup [44] and CutMix [48]), have proven to be effective to alleviate this problem. In this work, we integrate data augmentation into CL and propose an HCL module, which consists of supervised transformation-based CL and unsupervised mix-based CL, for better image representations.

For supervised CL, we use the scene class information to pull image features from the same class closer and push those from different classes away. Here, we adopt three data transformation strategies including random grayscale, color dithering, and level flipping, which basically do not change the image's semantic information and facilitate the supervised CL. Each time we randomly choose two of the three data transformation strategies to generate one transformed image, and we perform this operation twice on each input image so as to generate two augmented samples. As shown in Fig. 1, given a set of b images in a batch, we can obtain $2b$ augmented images denoted as $\{x_i\}_{i=1,\ldots,2b}$. We pass these augmented images to the weight-shared CNN encoders to obtain image representations, i.e.,

$$f_i = f_{\text{CNN}}(x_i) \tag{4}$$

where the CNN model can be the pretrained ResNet-18/34/50 or VGG-16. Then, we perform the MLP operation to obtain feature vectors, i.e.,

$$v_i = \text{MLP}(f_i) = \text{FC}(\text{ReLU}(\text{FC}(f_i))) \tag{5}$$

where MLP consists of two FCs and an ReLU. Finally, we adopt the following supervised transformation-based CL loss [45] to learn discriminative image representations:

$$\mathcal{L}_{\text{SupTrans}} = -\sum_{i=1}^{2b} \frac{1}{|P(i)|} \sum_{p\in P(i)} \log \frac{\exp\left(v_i \cdot v_p/\tau\right)}{\sum_{a\in A(i)} \exp\left(v_i \cdot v_a/\tau\right)} \tag{6}$$

where the index $i \in I \equiv \{1, \ldots, 2b\}$ indicates an anchor point, the index p indicates a positive sample, τ is a temperature coefficient, and $A(i) \equiv I \setminus \{i\}$. Here, $P(i) = \{p \in A(i) : y_p = y_i\}$ is a set of samples which have the same scene class as i, and $|P(i)|$ is the size of this set.

For unsupervised CL, we introduce CutMix [48] to randomly replace an image region with a patch from another training image to enhance image diversity. Compared with Mixup, CutMix can locally preserve image naturalness. Compared with transformation-based augmentation, CutMix can increase the image's semantic information and enables us to learn more discriminative scene representations for multilabel classification. As shown in Fig. 1, given b transformed images in a batch, we randomly select x_i and x_j from them to perform the following CutMix operation:

$$x_{ij}^{\text{mix}} = M \odot x_i + (1 - M) \odot x_j \tag{7}$$

where $M \in \{0, 1\}^{W\times H}$ denotes a binary mask indicating where to dropout and fill in from transformed images x_i and x_j, W and H are the width and height of images to be mixed, respectively, and $\odot$ denotes element-level multiplication. The binary mask M is randomly sampled as detailed in [48]. As done before, the mixed images x_{ij}^{mix} will also be fed into the weight-shared CNN encoders, followed by the MLP operation,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

to output feature vectors v_{ij}^{mix}. Finally, following [42] and [43], we define the unsupervised mix-based CL loss as

$$\mathcal{L}_{\text{UnsupMix}} = -\sum_{i=1}^{b}\left[\lambda_i \log\left(\frac{\exp\left(v_{i,j}^{\text{mix}}\cdot v_i/\tau\right)}{\sum_{k\in A_1(k)}\exp\left(v_{i,j}^{\text{mix}}\cdot v_k/\tau\right)}\right) + \left(1-\lambda_i\right)\log\left(\frac{\exp\left(v_{i,j}^{\text{mix}}\cdot v_j/\tau\right)}{\sum_{k\in A_2(k)}\exp\left(v_{i,j}^{\text{mix}}\cdot v_k/\tau\right)}\right)\right] \quad (8)$$

where the mixed image $x_{i,j}^{\text{mix}}$ is an anchor point with $i \in J \equiv \{1,\ldots,b\}$, v_i and v_j are the feature vectors of image x_i and x_j, respectively, obtained by (5), $A_1(k) \equiv J \setminus \{i\}$, $A_2(k) \equiv J \setminus \{j\}$, and λ_i is the area ratio of x_i in the mixed image $x_{i,j}^{\text{mix}}$. Note that $x_{i,j}^{\text{mix}}$ has semi-positive relation to x_i and x_j, which is reflected by the two loss items and their associated weights (λ_i and $1-\lambda_i$) in (8): the first one represents the similarity between $x_{i,j}^{\text{mix}}$ and x_i, and the second one represents the similarity between $x_{i,j}^{\text{mix}}$ and x_j.

To balance the supervised and unsupervised CL, we define the total loss of HCL as follows:

$$\mathcal{L}_{\text{HCL}} = \mathcal{L}_{\text{SupTrans}} + \alpha\mathcal{L}_{\text{UnsupMix}} \quad (9)$$

where α is a balance coefficient. As shown in Fig. 1, after the HCL module, the image representations of transformed images are sent to the multilabel classifier to obtain the initial label prediction scores through (3).

C. Scene-to-Label Prediction (SLP)

For RS images, scenes are closely related to object labels. Previous GCN-based methods only consider the relationship between labels, but neglect the guidance effects of scene information on label inference. To this end, we propose an SLP module using the scene-to-label information with the law of total probability to calibrate the initial label prediction scores obtained from the GCN classifier.

Formally, given an image representation f_i obtained by (4), we predict the probability that image x_i belongs to each specific scene class based on a softmax function

$$P_S = \text{Softmax}(\text{FC}(f_i)) \quad (10)$$

where $P_S = (p_{s_1},\ldots,p_{s_k},\ldots,p_{s_s})^T$ is a row vector of length s, s is the total number of scene classes in the training dataset, and p_{s_k} denotes the probability of image x_i belonging to the kth scene class, satisfying $\sum_{k=1}^{s} p_{s_k} = 1$.

Then, we compute the prior information of scene-to-label which is described as a conditional probability matrix

$$P_{L|S} = \left[P_{L|s_1};\ldots;P_{L|s_k};\ldots;P_{L|s_s}\right] \quad (11)$$

where $P_{L|s_k} = (p_{l_1|s_k},\ldots,p_{l_m|s_k},\ldots,p_{l_c|s_k})^T$ is a row vector of length c (c is the total number of object labels in the training dataset), and $P_{l_m|s_k}$ specifies the occurrence probability of the mth label when images belong to the kth scene class ($m = 1,\ldots,c$; $k = 1,\ldots,s$). The size of matrix $P_{L|S}$ is $s \times c$. Fig. 4 shows the scene-to-label probability matrix which is statistically obtained from the UCM dataset.

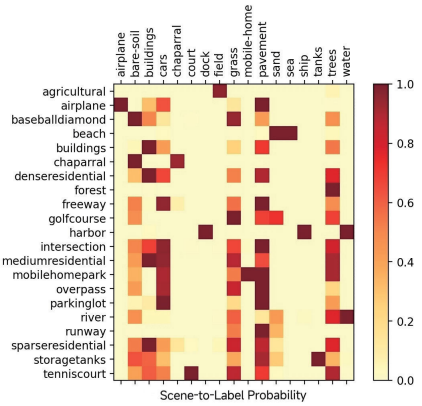


Fig. 4. Scene-to-label probability matrix for the UCM multilabel RS dataset. There are 21 scene classes and 17 object labels for this dataset, and hence the matrix is of size 21 × 17. This is a conditional probability matrix: e.g., the first row indicates that given an RS image belonging to the scene class "agricultural," it will contain the label "field" with a very high probability. The rest rows can be deduced by analogy.

According to the law of total probability, we can calculate the occurrence probabilities of all the labels in image x_i by multiplying P_S by $P_{L|S}$ in the following matrix form:

$$P_L = P_S \otimes P_{L|S} \quad (12)$$

where $P_L = (p_{l_1},\ldots,p_{l_m},\ldots,p_{l_c})^T$ is a row vector of length c, p_{l_m} denotes the occurrence probability of the mth label in image x_i, and $\otimes$ denotes matrix multiplication.

Finally, we use the derived label probabilities P_L to calibrate the initial label prediction scores obtained from the GCN classifier, i.e.,

$$\hat{y} = \hat{y}_0 \odot P_L^T \quad (13)$$

where $\hat{y}_0 \in \mathbb{R}^c$ is defined in (3), and $\odot$ denotes elementwise multiplication.

D. Multitask Learning With Uncertainty Loss

The proposed HSNet aims to learn discriminative scene representations and improve the performance of multilabel classification. For scene representations, we have designed the HCL module. To calibrate the initial multilabel prediction scores, we have designed the SLP module. Since the softmax function is adopted in SLP, we can readily use its output to compute the scene classification loss as a constraint condition to learn better scene representations. The final objective function of HSNet contains the following losses.

1) The HCL loss $\mathcal{L}_{\text{HCL}}$ defined in (9).
2) The scene classification loss

$$\mathcal{L}_{\text{scene}} = -\frac{1}{b}\sum_{i=1}^{b} z_i^T \log\left(\hat{z}_i\right) \quad (14)$$

where $z_i \in \mathbb{R}^s$ is the one-hot vector of the ith image's scene class (ground truth), $\hat{z}_i \in \mathbb{R}^s$ is the predicted probability distribution of scene class which is equal to $P_S{}^T$ in (10), and b is the batch size.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

3) The multilabel classification loss

$$\mathcal{L}_{\text{label}} = -\sum_{i=1}^{b}\sum_{j=1}^{c} y_{ij}\log\left(\hat{y}_{ij}\right) + \left(1-y_{ij}\right)\log\left(1-\hat{y}_{ij}\right) \tag{15}$$

where $y_{ij} = \{0, 1\}$ denotes the ground truth of the jth object label for image i, and $\hat{y}_{ij}$ is the predicted label value which is equal to the jth element of vector $\hat{y}$ defined in (13).

Typically, HSNet can perform multitask learning based on a naive weighted sum of the above losses. However, searching for the optimal weights is an expensive and difficult task with manual tuning. Inspired by [51], we use the uncertainty strategy to learn loss weighting and balance different tasks, which naturally lightens the manual tuning of weight hyperparameters. Specifically, by defining $\mathcal{L}_1 = \mathcal{L}_{\text{HCL}} + \beta\mathcal{L}_{\text{scene}} = \mathcal{L}_{\text{SupTrans}} + \alpha\mathcal{L}_{\text{UnsupMix}} + \beta\mathcal{L}_{\text{scene}}$ for scene representation and classification task, and $\mathcal{L}_2 = \mathcal{L}_{\text{label}}$ for multilabel classification task, the total uncertainty loss can be formulated as

$$\mathcal{L}_{\text{total}} = \frac{1}{2\sigma_1^2}\mathcal{L}_1 + \frac{1}{2\sigma_2^2}\mathcal{L}_2 + \log\sigma_1\sigma_2 \tag{16}$$

where σ_1 and σ_2 denote uncertain weights, which are initialized as 1 and then iteratively updated during the training phase. Hence, only the two hyperparameters α and β in $\mathcal{L}_{\text{total}}$ need to be determined.

IV. Experiments

A. Datasets

We evaluate HSNet for RS image multilabel classification on two publicly available datasets described as follows.

1) UCM Multilabel Dataset [52]: The UCM multilabel dataset is reproduced from the original UCM dataset [26] by reassigning images with multiple object labels. The dataset was collected from the United States Geological Survey (USGS) National Map. It contains 2100 RS images of 21 scene classes, and there are 100 256 × 256 RS images per class. The 21 scene classes are: agricultural, airplane, baseball-diamond, beach, buildings, chaparral, dense-residential, forest, freeway, golf-course, harbor, intersection, medium-residential, mobile home park, overpass, parking lot, river, runway, sparse-residential, storage-tanks, and tennis-court. The newly labeled UCM multilabel dataset has 17 object labels: airplane, bare-soil, buildings, cars, chaparral, court, dock, field, grass, mobile-home, pavement, sand, sea, ship, tanks, trees, and water. Fig. 5 shows some examples of this dataset with the corresponding scene class and labels.

2) Aerial Image Database (AID) Multilabel Dataset [14]: This dataset is recreated from the original AID dataset [53] by reassigning images with multiple object labels. The original AID dataset has 10 000 RGB images of 30 scene classes, and each class contains 220∼420 images with the spatial resolution 0.5∼8 m per pixel. The 30 scene classes are: airport, bare-land, baseball-field, beach, bridge, center, church, commercial, dense-residential, desert, farmland, forest, industrial, meadow, medium-residential, mountain, park, parking, playground, pond, port, railway-station, resort, river, school, sparse-residential, square, stadium, storage-tanks, and viaduct. Around 3000 images from 30 scenes are selected to construct the AID multilabel dataset, and the newly defined object labels are consistent with the UCM multilabel dataset. Fig. 6 shows some examples of this dataset with the corresponding scene class and labels.

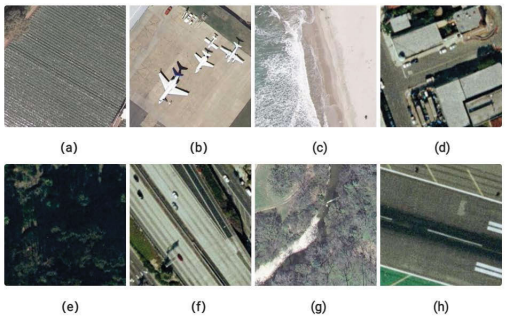


Fig. 5. Examples of the UCM multilabel dataset with the corresponding scene class and object labels. (a) Agricultural: field and trees. (b) Airplane: airplane, cars, grass, and pavement. (c) Beach: sand and sea. (d) Buildings: buildings, cars, and pavement. (e) Forest: trees. (f) Freeway: cars, grass, pavement, and trees. (g) River: bare-soil, grass, sand, trees, and water. (h) Runway: grass and pavement.

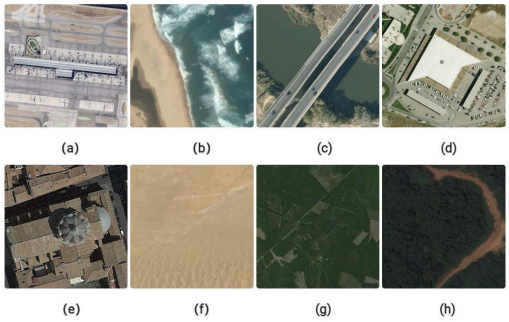


Fig. 6. Examples of the AID multilabel dataset with the corresponding scene class and object labels. (a) Airport: airplane, bare-soil, buildings, cars, grass, and pavement. (b) Beach: sand and sea. (c) Bridge: bare-soil, cars, grass, pavement, trees, and water. (d) Center: bare-soil, buildings, cars, chaparral, grass, pavement, and trees. (e) Church: buildings, cars, and pavement. (f) Desert: sand. (g) Farmland: field. (h) Forest: bare-soil and trees.

B. Experimental Setup

Following [6], [14], [17], and [21], we adopt average example- and label-based metrics to quantitatively evaluate the classification performance of HSNet. The example-based metrics include: 1) example-based F1 scores (F_e^1); 2) example-based F2 scores (F_e^2); 3) example-based precision (P_e); and 4) example-based recall (R_e). The label-based metrics include: 1) label-based F1 scores (F_l^1); 2) label-based F2 scores (F_l^2); 3) label-based precision (P_l); and 4) label-based recall (R_l). Generally, F1 and F2 scores are relatively important indicators for performance evaluation.

Specifically, the example-based F_e^β ($\beta = 1, 2$) score is defined as

$$F_e^\beta = \left(1+\beta^2\right)\frac{P_e R_e}{\beta^2 P_e + R_e} \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I

CLASSIFICATION PERFORMANCE IN TERMS OF FOUR EXAMPLE-BASED METRICS (%) ON THE UCM AND AID MULTILABEL DATASETS (THE BEST AND SECOND BEST RESULTS ARE IN BOLD AND UNDERLINED, RESPECTIVELY)

Method	Backbone	UCM				AID			
		F_e^1	F_e^2	P_e	R_e	F_e^1	F_e^2	P_e	R_e
ML-GCN [17]	ResNet-18	89.20	89.15	89.28	89.12	89.34	89.23	89.52	89.16
Ours		**90.62**	**90.46**	**90.89**	**90.35**	**90.54**	**90.24**	**91.04**	**90.04**
ML-GCN [17]	ResNet-34	88.15	87.75	88.83	87.48	87.45	87.97	86.59	88.32
Ours		**90.40**	**90.27**	**90.62**	**90.18**	**90.39**	**90.01**	**91.03**	**89.76**
ML-GCN [17]	ResNet-50	87.72	87.96	87.31	88.13	86.62	87.13	85.77	87.48
CA-BiLSTM [13]		83.11	86.56	77.94	89.02	-	-	-	-
SR-Net [15]		88.67	89.11	87.96	89.40	89.97	90.30	89.42	90.52
AL-RN [14]		87.93	87.41	88.81	87.07	89.96	89.35	**91.00**	88.95
ML-CG [21]		85.42	88.08	81.34	89.94	-	-	-	-
MSGM [22]		84.66	85.15	83.86	85.48	-	-	-	-
Ours		**90.53**	**90.63**	**90.37**	**90.69**	**90.36**	**90.85**	89.56	**91.18**
ML-GCN [17]	VGG-16	84.23	85.67	81.93	86.66	88.77	88.47	89.27	88.28
CA-BiLSTM [13]		81.59	83.01	79.33	83.99	-	-	-	-
SR-Net [15]		83.43	84.10	82.33	84.55	87.15	87.34	86.84	87.46
AL-RN [14]		87.01	86.65	87.62	86.41	89.61	89.41	89.96	89.27
MLRSSC-CNN-GNN [19]		87.76	88.15	87.11	88.41	90.01	90.13	89.83	90.20
Ours		**90.08**	**90.07**	**90.11**	**90.06**	**90.19**	**89.89**	**90.69**	**89.69**

where P_e and R_e are the example-based precision and recall, respectively, and they are calculated as follows:

$$P_e = \frac{1}{n}\sum_{i=1}^{n}\frac{\mathrm{TP}_e^i}{\mathrm{TP}_e^i + \mathrm{FP}_e^i} \tag{18}$$

$$R_e = \frac{1}{n}\sum_{i=1}^{n}\frac{\mathrm{TP}_e^i}{\mathrm{TP}_e^i + \mathrm{FN}_e^i} \tag{19}$$

where TP_e, FP_e, and FN_e represent the number of example-based true positive, false positive, and false negative, respectively, and n is the number of images in the test set.

Similarly, the label-based F_l^β ($\beta = 1, 2$) score is defined as

$$F_l^\beta = \left(1+\beta^2\right)\frac{P_l R_l}{\beta^2 P_l + R_l} \tag{20}$$

where P_l and R_l are the label-based precision and recall, respectively, and they are calculated as follows:

$$P_l = \frac{1}{c}\sum_{j=1}^{c}\frac{\mathrm{TP}_l^j}{\mathrm{TP}_l^j + \mathrm{FP}_l^j} \tag{21}$$

$$R_l = \frac{1}{c}\sum_{j=1}^{c}\frac{\mathrm{TP}_l^j}{\mathrm{TP}_l^j + \mathrm{FN}_l^j} \tag{22}$$

where TP_l, FP_l, and FN_l represent the number of label-based true positive, false positive, and false negative, respectively, and c is the number of labels in the test set.

We randomly select 70% of the images from each dataset for training, 10% for validation, and 20% for testing. We implement HSNet with the PyTorch framework and perform experiments on a workstation with an Intel Xeon W-2133 3.60-GHz processor and two NVIDIA GEFORCE RTX 2080 Ti 11-GB GPUs. In the training phase, we use three data transformation strategies, including random grayscale, color dithering, and level flipping, to generate transformed images. We train our model for 130 epochs with a batch size of 16, using Adam [54] with an initial learning rate of 1×10^{-5}, which is decayed by 0.5 every 30 epochs. We set $\alpha = 0.4$ and $\beta = 0.5$ in (16). We use four popular backbones as our CNN encoders for comprehensive evaluation: 1) ResNet-18 [31]; 2) ResNet-34 [31]; 3) ResNet-50 [31]; and 4) VGG-16 [29]. To compute evaluation metrics, we binarize the model's output with a threshold of 0.5 to generate final predictions.

C. Comparison With State-of-the-Art Methods

We compare the proposed HSNet with the state-of-the-art multilabel classification approaches which are detailed below.

1) *ML-GCN [17]:* A directed label graph is constructed based on word embeddings and a co-occurrence matrix to output a multilabel classifier.
2) *CA-BiLSTM [13]:* A bidirectional LSTM-based subnetwork is designed to model the underlying class dependency and predict multiple object labels.
3) *MLRSSC-CNN-GNN [19]:* A scene graph is constructed where nodes of the graph are represented by superpixel regions and a multilayer integration graph attention network is designed for multilabel classification.
4) *SR-Net [15]:* The dependency relationship between labels is modeled by a transformer-driven semantic relation inference network.
5) *AL-RN [14]:* The dependency relationship between labels is modeled by an attention-aware label relational reasoning network.
6) *ML-CG [21]:* Label correlations are inferred from both the specific dataset and ConceptNet. A concept attention graph-neural-network-based framework is proposed for multilabel classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

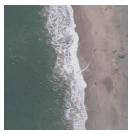
TABLE II

CLASSIFICATION PERFORMANCE IN TERMS OF FOUR LABEL-BASED METRICS (%) ON THE UCM AND AID MULTILABEL DATASETS (THE BEST AND SECOND BEST RESULTS ARE IN BOLD AND UNDERLINED, RESPECTIVELY)

Method	Backbone	UCM				AID			
		F_l^1	F_l^2	P_l	R_l	F_l^1	F_l^2	P_l	R_l
ML-GCN [17]	ResNet-18	88.10	88.37	87.65	88.55	88.26	88.49	87.88	88.64
Ours		**89.62**	**89.78**	**89.35**	**89.88**	**90.02**	**89.64**	**90.66**	**89.39**
ML-GCN [17]	ResNet-34	87.81	88.46	86.75	88.90	86.00	87.00	84.38	87.68
Ours		**89.54**	**89.93**	**88.90**	**90.19**	**89.28**	**89.75**	**88.51**	**90.07**
ML-GCN [17]	ResNet-50	86.21	87.25	84.53	87.96	83.25	82.71	82.98	82.81
CA-BiLSTM [13]		85.18	84.63	86.12	84.26	-	-	-	-
SR-Net [15]		**92.50**	**91.91**	**93.52**	**91.51**	84.67	83.20	87.24	82.25
AL-RN [14]		89.02	87.15	92.33	85.95	75.66	72.87	80.81	71.12
ML-CG [21]		88.90	89.12	88.53	89.27	-	-	-	-
MSGM [22]		87.46	86.01	89.98	85.07	-	-	-	-
Ours		89.46	89.88	88.77	90.16	**89.35**	**89.48**	**89.13**	**89.57**
ML-GCN [17]	VGG-16	83.21	82.36	82.78	82.52	84.45	84.68	84.56	84.63
CA-BiLSTM [13]		80.66	78.13	85.28	76.52	-	-	-	-
SR-Net [15]		83.67	81.85	86.90	80.68	70.20	67.23	75.79	65.38
AL-RN [14]		86.12	83.42	**91.04**	81.71	72.37	69.88	76.94	68.31
Ours		**89.32**	**89.53**	88.97	**89.67**	**89.26**	**89.46**	**88.93**	**89.59**

TABLE III

EXAMPLE PREDICTIONS ON THE UCM AND AID MULTILABEL DATASETS

Samples from the UCM multilabel data set					
Ground-truth labels	airplane, grass, pavement	sand, sea	buildings, trees	grass, sand, trees, water	cars, pavement
Predicted labels	airplane, grass, pavement	sand, sea	buildings, trees	grass, sand, trees, water,	cars, pavement, grass
Samples from the AID multilabel data set					
Ground-truth labels	bare-soil, buildings, cars, field, pavement, trees	bare-soil, buildings, cars, court, grass, pavement, trees, water	buildings, court, grass, pavement, trees, water	bare-soil, buildings, cars, court, grass, pavement, trees	buildings, cars, pavement, trees
Predicted labels	bare-soil, buildings, cars, field, pavement, trees	bare-soil, buildings, cars, court, grass, pavement, trees, water	buildings, court, grass, pavement, trees, water, bare-soil	bare-soil, buildings, cars, court, grass, pavement, trees	buildings, cars, pavement, trees

Notes: Red predictions are false positives, and blue predictions are false negatives.

7) *MSGM [22]:* A multigranularity semantic grouping model is built by combining grouping features and GCN.

We implement ML-GCN by ourselves as the baseline based on the published codes. The results of other compared methods are taken from the original articles. Tables I and II show the classification results of different methods on the UCM and AID multilabel datasets in terms of example-based and label-based metrics, respectively. We can make the following observations.

1) First, without using graph convolution, CA-BiLSTM uses CNN-LSTM to model the underlying class dependencies and generally performs poorly on the UCM dataset in terms of all metrics.
2) Second, MSGM using feature grouping and graph convolution outperforms CA-BiLSTM in terms of label-based metrics on the UCM dataset, but performs worse than CA-BiLSTM in terms of F_e^2 and R_e metrics. In addition, ML-GCN using GCNs and ML-CG using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
EFFECTIVENESS OF DIFFERENT COMPONENTS USED IN HSNET IN TERMS OF FOUR EXAMPLE-BASED METRICS

	Components				UCM				AID			
	SLP	$\mathcal{L}_{SupTrans}$	$\mathcal{L}_{UnsupMix}$	$\mathcal{L}_{scene}$	F_e^1	F_e^2	P_e	R_e	F_e^1	F_e^2	P_e	R_e
Baseline (ML-GCN)					89.20	89.15	89.28	89.12	89.34	89.23	89.52	89.16
+ SLP + $\mathcal{L}_{scene}$	✓			✓	89.54	89.63	89.39	89.69	89.47	89.36	89.65	89.29
+ $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{scene}$		✓		✓	89.42	89.50	89.29	89.55	89.38	89.29	89.53	89.23
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{scene}$	✓	✓		✓	89.78	89.98	89.45	90.11	89.68	89.55	89.90	89.47
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{UnsupMix}$	✓	✓	✓		90.19	90.23	90.13	90.26	90.08	90.01	90.20	89.96
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{UnsupMix}$ + $\mathcal{L}_{scene}$	✓	✓	✓	✓	**90.62**	**90.46**	**90.89**	**90.35**	**90.54**	**90.24**	**91.04**	**90.04**

TABLE V
EFFECTIVENESS OF DIFFERENT COMPONENTS USED IN HSNET IN TERMS OF FOUR LABEL-BASED METRICS

	Components				UCM				AID			
	SLP	$\mathcal{L}_{SupTrans}$	$\mathcal{L}_{UnsupMix}$	$\mathcal{L}_{scene}$	F_l^1	F_l^2	P_l	R_l	F_l^1	F_l^2	P_l	R_l
Baseline (ML-GCN)					88.10	88.37	87.65	88.55	88.26	88.49	87.88	88.64
+ SLP + $\mathcal{L}_{scene}$	✓			✓	88.62	88.78	88.35	88.89	88.69	88.70	88.67	88.71
+ $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{scene}$		✓		✓	88.57	88.65	88.44	88.70	88.42	88.58	88.16	88.69
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{scene}$	✓	✓		✓	89.13	89.08	89.21	89.05	89.15	89.05	89.32	88.98
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{UnsupMix}$	✓	✓	✓		89.18	89.14	89.24	89.12	89.37	89.17	89.71	89.03
+ SLP + $\mathcal{L}_{SupTrans}$ + $\mathcal{L}_{UnsupMix}$ + $\mathcal{L}_{scene}$	✓	✓	✓	✓	**89.62**	**89.78**	**89.35**	**89.88**	**90.02**	**89.64**	**90.66**	**89.39**

concept graph outperform CA-BiLSTM in terms of F1 and F2 scores, demonstrating the effectiveness of graph convolution for label modeling.

3) Third, MLRSSC-CNN-GNN (VGG-16) uses superpixels to construct graph convolution, which achieves the second best results in terms of example-based metrics on both the datasets.
4) Fourth, the proposed model performs best overall on both the datasets. It is worth mentioning that SR-Net using ResNet-50 as the backbone performs best on the UCM dataset in terms of four label-based metrics, but it is not the case on the AID dataset or in terms of other metrics.
5) Finally, Table III shows some prediction results for images from the UCM and AID multilabel datasets. We can see that the easily confused appearance and color changes account for these wrongly predicted labels.

D. Analysis of HSNet

Now we analyze HSNet from different aspects and use ResNet-18 as the CNN encoder by default.

1) Effectiveness of Key Components: Built upon ML-GCN, the proposed HSNet contains two novel modules, namely, HCL and SLP, which involve three key loss functions: $\mathcal{L}_{\text{SupTrans}}$, $\mathcal{L}_{\text{UnsupMix}}$, and $\mathcal{L}_{\text{scene}}$. We evaluate the effectiveness of these components and present the results in Tables IV and V (note that the compared schemes except ML-GCN have used the uncertainty strategy to handle multiple losses). By comparing the results in Rows 4 and 2 (Row 3), we can see the effectiveness of $\mathcal{L}_{\text{SupTrans}}$ (SLP). By comparing the results in Rows 6 and Row 4 (Row 5), we can see the effectiveness of $\mathcal{L}_{\text{UnsupMix}}$ ($\mathcal{L}_{\text{scene}}$). Hence, the HCL and SLP modules together with the corresponding loss functions are all effective for multilabel classification.

TABLE VI
EFFECTIVENESS OF THE UNCERTAINTY STRATEGY USED IN HSNET IN TERMS OF FOUR EXAMPLE-BASED METRICS

Uncertainty	UCM				AID			
	F_e^1	F_e^2	P_e	R_e	F_e^1	F_e^2	P_e	R_e
✗	90.12	90.43	89.64	90.26	90.03	89.56	90.81	89.26
✓	**90.62**	**90.46**	**90.89**	**90.35**	**90.54**	**90.24**	**91.04**	**90.04**

2) Effectiveness of Uncertainty Strategy: The uncertainty strategy is used in HSNet to learn loss weighting and balance different tasks. As shown in Tables VI and VII, the use of uncertainty loss can improve the classification results in terms of both example-based and label-based metrics. This is because the uncertainty loss enables the model to learn to optimize multitask weights, which promotes the interaction between different tasks. In addition, the uncertainty loss lightens the manual tuning of weight hyperparameters.

3) Effects of Feature Dimensions of Word Vectors: The word vectors of labels used in HSNet are obtained from GloVe trained on the Wikipedia data. We evaluate the effects of feature dimensions of word vectors on the multilabel classification results. As shown in Tables VIII and IX, the use of 300-D word vectors leads to the best performance in terms of all the metrics. Hence, we set the dimensions of word vectors to be 300 by default in the subsequent experiments.

4) Comparison of Mixup and CutMix: The proposed HSNet has used CutMix, instead of Mixup, in the HCL module to generate diverse images while preserving local naturalness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

EFFECTIVENESS OF THE UNCERTAINTY STRATEGY USED IN HSNET IN TERMS OF FOUR LABEL-BASED METRICS

Uncertainty	UCM				AID			
	F_l^1	F_l^2	P_l	R_l	F_l^1	F_l^2	P_l	R_l
✗	88.89	89.57	88.71	89.06	88.79	88.85	88.70	88.89
✓	**89.62**	**89.78**	**89.35**	**89.88**	**90.02**	**89.64**	**90.66**	**89.39**

TABLE VIII

CLASSIFICATION RESULTS IN TERMS OF FOUR EXAMPLE-BASED METRICS BY VARYING THE FEATURE DIMENSIONS (DIMS.) OF WORD VECTORS

Dims.	UCM				AID			
	F_e^1	F_e^2	P_e	R_e	F_e^1	F_e^2	P_e	R_e
50	89.92	89.93	89.90	89.95	89.51	89.43	89.66	89.37
100	89.40	89.78	88.76	90.04	90.06	89.97	90.22	89.91
200	89.15	89.51	88.54	89.76	89.80	89.16	90.88	88.74
300	**90.62**	**90.46**	**90.89**	**90.35**	**90.54**	**90.24**	**91.04**	**90.04**

TABLE IX

CLASSIFICATION RESULTS IN TERMS OF FOUR LABEL-BASED METRICS BY VARYING THE FEATURE DIMENSIONS (DIMS.) OF WORD VECTORS

Dims.	UCM				AID			
	F_l^1	F_l^2	P_l	R_l	F_l^1	F_l^2	P_l	R_l
50	88.80	89.20	88.14	89.47	88.91	88.35	89.86	87.98
100	89.36	89.46	89.21	89.52	88.95	88.10	90.40	87.54
200	89.29	89.36	89.17	89.41	89.52	89.36	89.79	89.25
300	**89.62**	**89.78**	**89.35**	**89.88**	**90.02**	**89.64**	**90.66**	**89.39**

TABLE X

PERFORMANCE COMPARISON FOR MIXUP AND CUTMIX IN TERMS OF FOUR EXAMPLE-BASED METRICS

	UCM				AID			
	F_e^1	F_e^2	P_e	R_e	F_e^1	F_e^2	P_e	R_e
Mixup	90.57	90.43	90.50	**90.46**	90.38	90.20	90.68	**90.08**
CutMix	**90.62**	**90.46**	**90.89**	90.35	**90.54**	**90.24**	**91.04**	90.04

TABLE XI

PERFORMANCE COMPARISON FOR MIXUP AND CUTMIX IN TERMS OF FOUR LABEL-BASED METRICS

	UCM				AID			
	F_l^1	F_l^2	P_l	R_l	F_l^1	F_l^2	P_l	R_l
Mixup	89.54	89.69	89.29	89.79	89.91	89.56	90.50	89.33
CutMix	**89.62**	**89.78**	**89.35**	**89.88**	**90.02**	**89.64**	**90.66**	**89.39**

Tables X and XI show comparison of the classification results in terms of example- and label-based metrics, respectively. Overall, CutMix outperforms Mixup in terms of all the metrics. This is because the samples generated by Mixup are usually visually misty and locally unnatural. In contrast, CutMix generates diverse samples in a cut-and-paste manner which can preserve local image naturalness. Hence, CutMix helps learn discriminative image representations for multilabel classification.

TABLE XII

SENSITIVITY ANALYSIS OF HYPERPARAMETERS α AND β IN TERMS OF F_l^1 VALUE ON THE UCM (COLUMNS 2–6) AND AID (COLUMNS 7–11) DATASETS

α \ β	0.1	0.3	0.5	0.8	1.0	0.1	0.3	0.5	0.8	1.0
1.0	89.16	89.34	89.23	89.16	89.12	89.28	89.48	89.26	89.13	89.06
0.7	89.36	89.41	89.52	89.43	89.21	89.61	89.75	89.58	89.37	89.90
0.4	89.25	89.59	**89.62**	89.43	88.94	89.35	89.96	**90.02**	89.38	88.89
0.1	89.20	88.68	88.51	89.18	88.26	89.63	89.45	89.26	89.44	89.01

TABLE XIII

COMPARISON OF THE NUMBER OF PARAMETERS, FLOPS, AND THE TESTING TIME PER SAMPLE

Method	Backbone	#Parameters (M)	#FLOPs (G)	Testing Time (ms)
ML-GCN	ResNet-18	11.385	2.224	4.681
	ResNet-34	21.493	4.482	7.986
	ResNet-50	25.912	5.078	10.694
	VGG-16	14.931	18.754	5.355
Ours	ResNet-18	11.669	2.225	4.912
	ResNet-34	21.777	4.483	8.572
	ResNet-50	30.195	5.079	11.531
	VGG-16	15.215	18.755	5.742

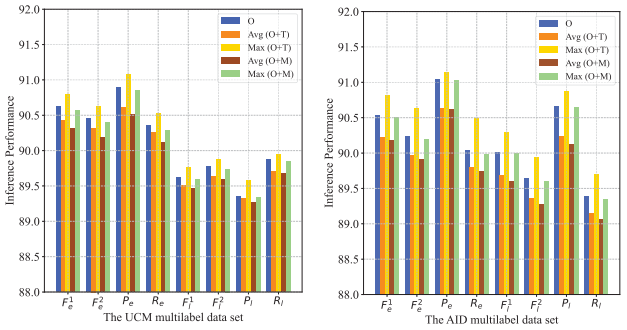


Fig. 7. Classification performance obtained using different data augmentation schemes during inference on the UCM and AID multilabel datasets.

5) Parameter Sensitivity Analysis: Two hyperparameters α and β in the total uncertainty loss remain to be determined, which may affect the final classification performance. We evaluate their effects by varying their values from 0.1 to 1 and briefly present the results in terms of F_l^1 in Table XII. One can see that when $\alpha = 0.4$ and $\beta = 0.5$, HSNet achieves the best classification results on both the datasets. In fact, by observing Table II, we find that HSNet always outperforms the baseline ML-GCN when α and β are varied from 0.1 to 1. This demonstrates the efficacy of our multitask learning and optimization framework.

6) Complexity Analysis: We analyze the complexity of ML-GCN and the proposed HSNet by comparing the number of model parameters, FLOPs, and the testing time per sample. As shown in Table XIII, compared with ML-GCN, HSNet requires slightly increased costs in the number of parameters, FLOPs, and the testing time for processing each sample. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

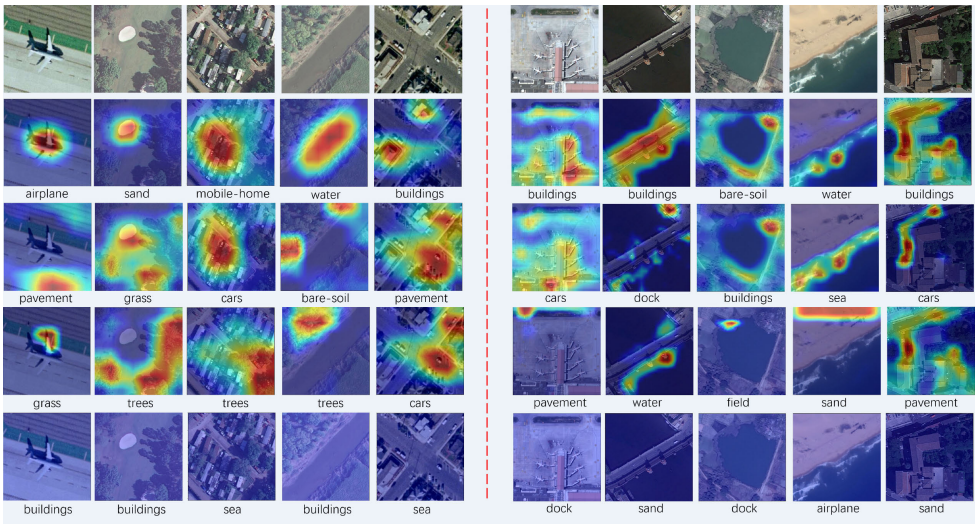


Fig. 8. Some example images and the corresponding label activation maps which highlight label-specific discriminative regions. (Left) UCM multilabel dataset. (Right) AID multilabel dataset. Row 1: the original image and Rows 2–5: the label activation maps for specific object labels (the first three object labels in each column are true while the last one is not true).

is because HSNet additionally uses a fully connected layer and some multiplication operations (Fig. 1). Overall, HSNet can achieve good classification performance with relative low computational and storage costs.

7) Data Augmentation During Inference: For the proposed HSNet, we have used both transformed and mixed (via CutMix) images to train our network, while we only used the one-branch transformed image corresponding to f_i in Fig. 1 for inference (we denote this original scheme as "O"). As alternative inference schemes, we can use 1) two-branch transformed images (denoted as "O + T," corresponding to f_i and f_k), or 2) one-branch transformed image plus one-branch mixed image (denoted as "O + M," corresponding to f_i and f_{ij}^{mix}) to compute image representations and take the maximum ("Max") or average ("Avg") value of two branches as the final prediction score for each label. Fig. 7 shows the classification performance obtained using the above different schemes on the UCM and AID datasets. We can see that the "Max" scheme outperforms the "Avg" scheme, and "Max" (O + T) outperforms "Max" (O + M). Using two-branch transformed images, "Max" (O + T) achieves the best results on both the datasets in terms of all the metrics at the cost of more inference time compared with the original HSNet (i.e., "O"). In contrast, "Max" (O + M) shows poorer performance than "O." This is because the mixed (via CutMix) images can introduce external and unrelated image patches, hence changing the semantic information of original input images and affecting the label's prediction scores.

8) Visualization Results: In Fig. 8, we visualize some example images and the corresponding label activation maps via Grad-CAM [55]. Note that with ResNet-18 as the backbone, Grad-CAM is used to activate the last convolutional layer of ResNet-18. As shown in Fig. 8, the semantic regions corresponding to true object labels present in the image are correctly activated, while those corresponding to nonexistent labels in the image are not activated (the object labels in Row 5 are not true and they are not activated). These results indicate that the proposed network can well capture discriminative semantic information for multilabel RS image classification.

V. Conclusion

We propose HSNet, a GCN-based deep network model for multilabel RS image classification, which incorporates CL, scene classification and multilabel classification into a unified learning framework. Specifically, we introduce an HCL module to explicitly model scene representations based on supervised transformation-based CL and unsupervised mix-based CL. We design an SLP module to predict the scene probability and explore the scene-to-label information to calibrate the initial label prediction scores obtained from the GCN classifier. We take advantage of uncertainty to learn loss weighting and balance different tasks. The experimental results on two RS datasets demonstrate the effectiveness of HSNet for multilabel image classification in terms of example-based and label-based metrics.

HSNet has some possible limitations that may be addressed in the future work. For instance, HSNet does not explicitly consider the scene information to construct the GCN and relies only on the subsequent SLP module to calibrate the initial scores. An alternative solution may be to directly incorporate the scene information into GCNs. In addition, HSNet explores both visual and text information for multilabel classification. As an extension of our method, we can further take advantage of multisource data [56], [57] for multilabel image classification, especially using the hyperspectral images which contain rich spectral information to identify different land cover types and predict the associated semantic labels more accurately.

References

[1] H. Huang, G. Shi, H. He, Y. Duan, and F. Luo, "Dimensionality reduction of hyperspectral imagery based on spatial–spectral manifold learning," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2604–2616, Jun. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

[2] W. Zhang, X. Lu, and X. Li, "A coarse-to-fine semi-supervised change detection for multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3587–3599, Jun. 2018.

[3] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.

[4] X. Lu, X. Zheng, and Y. Yuan, "Remote sensing scene classification by unsupervised representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5148–5157, Sep. 2017.

[5] G. Cheng et al., "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608011.

[6] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, Aug. 2014.

[7] J. Wang, Y. Yang, J. Mao, Z. Huang, C. Huang, and W. Xu, "CNN-RNN: A unified framework for multi-label image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2285–2294.

[8] F. Liu, T. Xiang, T. M. Hospedales, W. Yang, and C. Sun, "Semantic regularisation for recurrent image annotation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4160–4168.

[9] J. Zhang, J. Ren, Q. Zhang, J. Liu, and X. Jiang, "Spatial context-aware object-attentional network for multi-label image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 3000–3012, 2023.

[10] J. Zhang, Q. Wu, C. Shen, J. Zhang, and J. Lu, "Multilabel image classification with regional latent semantic dependencies," *IEEE Trans. Multimedia*, vol. 20, no. 10, pp. 2801–2813, Oct. 2018.

[11] Z. Wang, T. Chen, G. Li, R. Xu, and L. Lin, "Multi-label image recognition by recurrently discovering attentional regions," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 464–472.

[12] F. Zhu, H. Li, W. Ouyang, N. Yu, and X. Wang, "Learning spatial regularization with image-level supervisions for multi-label image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5513–5522.

[13] Y. Hua, L. Mou, and X. X. Zhu, "Recurrently exploring class-wise attention in a hybrid convolutional and bidirectional LSTM network for multi-label aerial image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 188–199, Mar. 2019.

[14] Y. Hua, L. Mou, and X. X. Zhu, "Relation network for multilabel aerial image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4558–4572, Jul. 2020.

[15] X. Tan, Z. Xiao, J. Zhu, Q. Wan, K. Wang, and D. Li, "Transformer-driven semantic relation inference for multilabel classification of high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1884–1901, 2022.

[16] L. Bashmal, Y. Bazi, M. M. A. Rahhal, H. Alhichri, and N. A. Ajlan, "UAV image multi-labeling with data-efficient transformers," *Appl. Sci.*, vol. 11, no. 9, p. 3974, Apr. 2021.

[17] Z. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5172–5181.

[18] T. Chen, M. Xu, X. Hui, H. Wu, and L. Lin, "Learning semantic-specific graph representation for multi-label image recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 522–531.

[19] Y. Li, R. Chen, Y. Zhang, M. Zhang, and L. Chen, "Multi-label remote sensing image scene classification by combining a convolutional neural network and a graph neural network," *Remote Sens.*, vol. 12, no. 23, p. 4003, Dec. 2020.

[20] J. Kang, R. Fernandez-Beltran, D. Hong, J. Chanussot, and A. Plaza, "Graph relation network: Modeling relations between scenes for multilabel remote-sensing image classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4355–4369, May 2021.

[21] D. Lin, J. Lin, L. Zhao, Z. J. Wang, and Z. Chen, "Multilabel aerial image classification with a concept attention graph neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602112.

[22] D. Lin and Z. Chen, "Semantic understandings for aerial images via multigrained feature grouping," *Sci. Program.*, vol. 2022, pp. 1–12, Apr. 2022.

[23] Q. Tan, Y. Liu, X. Chen, and G. Yu, "Multi-label classification based on low rank representation for image annotation," *Remote Sens.*, vol. 9, no. 2, p. 109, Jan. 2017.

[24] Y. Yang and S. Newsam, "Comparing SIFT descriptors and Gabor texture features for classification of remote sensed imagery," in *Proc. 15th IEEE Int. Conf. Image Process.*, Oct. 2008, pp. 1852–1855.

[25] R. M. Anwer, F. S. Khan, J. van de Weijer, M. Molinier, and J. Laaksonen, "Binary patterns encoded convolutional neural networks for texture recognition and remote sensing scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 74–85, Apr. 2018.

[26] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[27] B. Zhao, Y. Zhong, G. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016.

[28] Y. Yuan, J. Fang, X. Lu, and Y. Feng, "Remote sensing image scene classification using rearranged local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1779–1792, Mar. 2019.

[29] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Intl. Conf. Learning Represent. (ICLR)*, 2015, pp. 1–14.

[30] H. Zheng, T. Song, C. Gao, and T. Guo, "M2FN: A multilayer and multi-attention fusion network for remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[32] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[33] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[34] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.

[35] A. Zeggada, F. Melgani, and Y. Bazi, "A deep learning approach to UAV image multilabeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 694–698, May 2017.

[36] S. Koda, A. Zeggada, F. Melgani, and R. Nishii, "Spatial and structured SVM for multilabel image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5948–5960, Oct. 2018.

[37] R. Stivaktakis, G. Tsagkatakis, and P. Tsakalides, "Deep learning for multilabel land cover scene categorization using data augmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1031–1035, Jul. 2019.

[38] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[39] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[40] T. Chen, S. Kornblith, K. Swersky, M. Norouzi, and G. E. Hinton, "Big self-supervised models are strong semi-supervised learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 22243–22255.

[41] H. Li et al., "Global and local contrastive self-supervised learning for semantic segmentation of HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618014.

[42] S. Kim, G. Lee, S. Bae, and S.-Y. Yun, "MixCo: Mix-up contrastive learning for visual representation," 2020, *arXiv:2010.06300*.

[43] K. Wickstrøm, M. Kampffmeyer, K. Ø. Mikalsen, and R. Jenssen, "Mixing up contrastive learning: Self-supervised representation learning for time series," *Pattern Recognit. Lett.*, vol. 155, pp. 54–61, Mar. 2022.

[44] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–13.

[45] P. Khosla et al., "Supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 18661–18673.

[46] M. Huang, L. Dong, W. Dong, and G. Shi, "Supervised contrastive learning based on fusion of global and local features for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5208513.

[47] J. Cui, Z. Zhong, S. Liu, B. Yu, and J. Jia, "Parametric contrastive learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 695–704.

[48] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.

[49] J. Pennington, R. Socher, and C. Manning, "Glove: Global vectors for word representation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1532–1543.
[50] A. L. Maas, "Rectifier nonlinearities improve neural network acoustic models," in *Proc. 30th Int. Conf. Mach. Learn. (ICML)*, Jun. 2013, pp. 1–6.
[51] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7482–7491.
[52] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.
[53] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–13.
[55] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.
[56] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.
[57] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.



Tiecheng Song (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2015.

From October 2015 to April 2016, he was a Visiting Student with the Multimedia Laboratory, Nanyang Technological University, Singapore. He is currently a Professor with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. His research interests include image processing and computer vision.



Shufen Bai received the B.S. degree in information engineering from Hunan University of Technology, Zhuzhou, China, in 2021. She is currently pursuing the M.S. degree with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China.

Her research interests include remote image processing and deep learning.



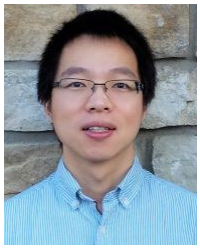
Feng Yang received the B.S. degree in science and technology of remote sensing and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013 and 2018, respectively.

She is currently an Associate Professor with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. Her research interests include remote sensing imaging, video processing, and computer vision.



Chenqiang Gao received the B.S. degree in computer science from China University of Geosciences, Wuhan, China, in 2004, and the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, in 2009.

In August 2009, he joined the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. In September 2012, he joined the Informedia Group, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, USA, where he was a Visiting Scholar on multimedia event detection (MED) and surveillance event detection (SED). In April 2013, he became a Post-Doctoral Fellow and continued work on MED and SED until March 2014, when he returned to CQUPT. In September 2023, he joined the School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, Guangdong, China. His research interests include image processing, infrared target detection, action recognition, and event detection.



Haonan Chen (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from Colorado State University (CSU), Fort Collins, CO, USA, in 2017.

He has been an Assistant Professor with the Department of Electrical and Computer Engineering, CSU, since August 2020. He is currently an Affiliate Faculty with the Data Science Research Institute (DSRI), CSU. Before joining the CSU Faculty, he was with the Cooperative Institute for Research in the Atmosphere (CIRA), Fort Collins, and the National Oceanic and Atmospheric Administration (NOAA) Physical Sciences Laboratory, Boulder, CO, USA, from 2012 to 2020, first as a Research Student, then a National Research Council Research Associate, and a Radar, Satellite, and Precipitation Research Scientist. His research interests include remote sensing and multidisciplinary data science, including radar and satellite remote sensing of natural disasters, polarimetric radar systems and networking, clouds and precipitation observations and processes, big data analytics, and deep learning.

Dr. Chen serves as an Associate Editor for the *Journal of Atmospheric and Oceanic Technology*, URSI Radio Science Bulletin, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisbon, Lisbon, Portugal, in 2011.

She is currently a Full Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. Her research interests include remotely sensed hyperspectral image analysis, signal processing, supervised and unsupervised learning, and active learning.

Prof. Li is the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been the Guest Editor of several journals, including PROCEEDINGS OF THE IEEE and *ISPRS Journal of Photogrammetry and Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:04:40 UTC from IEEE Xplore. Restrictions apply.